

Rhidmo® Installation Manual

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1 What is Rhidmo?

1.1 Introduction

1.2 Prerequisites


```

>deploy /tmp/deploy/rhidmo-ear-1.0.ear

Deploy settings:
  life_cycle=bulk
  on_deploy_error=stop
  on_prerequisite_error=stop
  version_rule=lower
  workflow=normal

If there is an offline deployment, Telnet connection to host may be lost, but the
result can be seen using [get_result] command

Processing deployment operation, wait...

===== PROGRESS START =====

Deploying [de.foxysoft_rhidmo (sda)] ...
Deployment of [de.foxysoft_rhidmo (sda)] finished.

===== PROGRESS END =====

===== DEPLOY RESULT =====

  sdu id: [de.foxysoft_rhidmo]
sdu file path: [/usr/sap/1117/J00/j2ee/cluster/server0/temp/tc-bl-deploy_controller/archives/131/12311243856035/rhidmo-ear-1.0.ear]
version status: [HIGHER]
deployment status: [Success]
description: []

===== END DEPLOY RESULT =====

===== Summary - Deploy Result - Start =====
-----
Type | Status : Count
-----
> SCA(s)
> SDA(s)
  - [Success] : [1]
-----
Type | Status : Id
-----
> SCA(s)
> SDA(s)
  - [Success] : de.foxysoft_rhidmo,
-----
===== Summary - Deploy Result - End =====

>

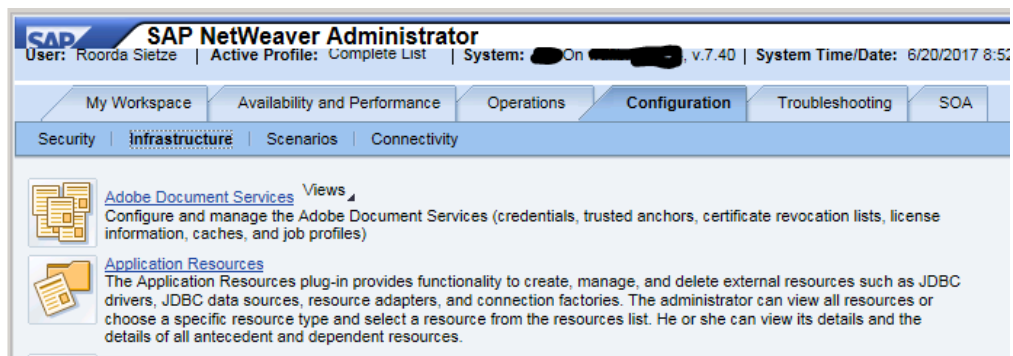
```

Watch for the “Success” message (as shown in the screenshot above). Correct any errors. Rhidmo is now deployed and must be configured before use. This is explained in the following chapter.

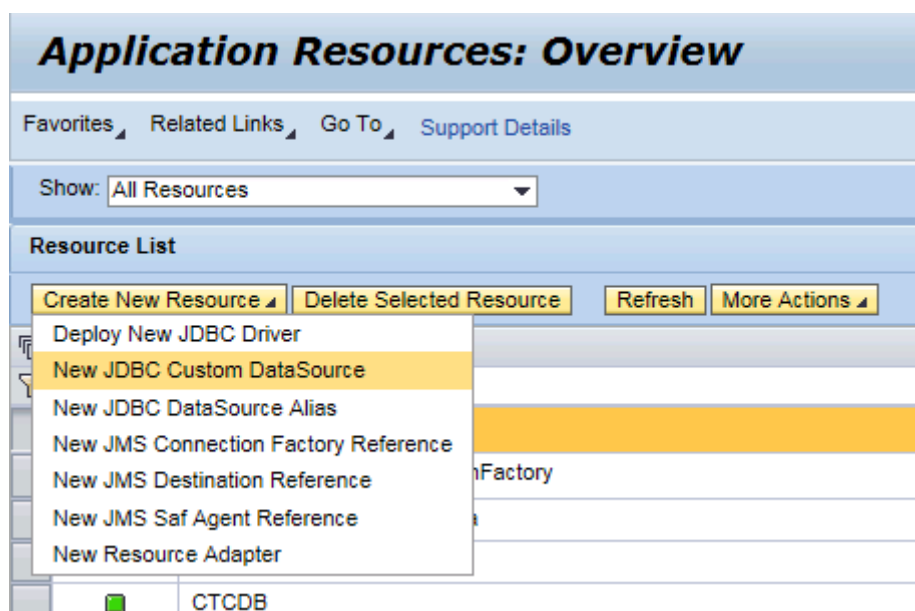
2.2 Create RHIDMO_RT Datasource

The standard IDM data source that is configured in JAVA AS uses the “<prefix>_prov” user which does not have sufficient privileges to be used with Rhidmo. Consequently, another data source must be configured with the “<prefix>_rt” user:

1. Log in with a SAP supported browser to the JAVA AS with a user with administrative privileges and start the NetWeaver Administrator (URL: <http://<machine>:<port>/nwa>). Use https if available:



2. Go to “Configuration” → “Infrastructure” (as shown in the screenshot above) and click on the “Application Resources” link. Press the button “New JDBC Custom DataSource”:



3. Fill in the form with the same values as for the “IDM_DataSource” Data Source. Please remember to exchange the “<prefix>_prov” user with the “<prefix>_rt” user (and adjust the password accordingly):

New JDBC Custom DataSource Creation

Save Cancel

Settings Connection Pooling Additional Properties JDBC DataSource Aliases

Application Name: RHIDMO_RT

DataSource Name: * RHIDMO_RT

Driver Name: * MSS

SQL Engine: * Native SQL

Isolation Level: * Default

JDBC Version: * 1x (without XA support)

Driver Class Name: * com.microsoft.sqlserver.jdbc.SQLServerDriver

Database URL: * jdbc:sqlserver://192.168.7.61:1433;databasename=idm80_db

User Name: * idm80_rt

Password: *

Description: Rhidmo DataSource

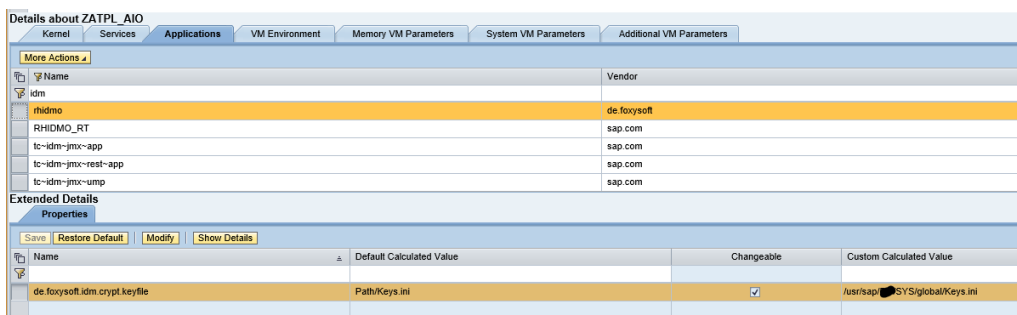
The screenshot above shows example values for the Microsoft SQL Server database using “idm80” as the prefix.

2.3 Set Rhidmo Application Parameters

1. While still in NWA, go to “Configuration” → “Infrastructure”:



2. Press the link “Java System Properties” and find the tab “Applications” (somewhere in the middle). Filter on “idm” to see only SAP IDM relevant content:



3. Change the property “de.foxysoft.idm.crypt.file” to have the same value as the property “com.sap.idm.jmx.crypt.keyfile” of the “tc~idm~jmx~app” application. Press the “Save” button to store your changes.

3 Rhidmo Example Code

Starting with release 1.2.0, Rhidmo includes demo packages with example code which can be used after the Rhidmo EAR has been deployed. While none of this content is recommended or even required for production use, developers may use it as a helpful starting point to learn more about how to use Rhidmo. Please refer also to chapter 3.6 for more developer-oriented material. Finally, the demo might also be useful as a ready-to-use example of Rhidmo in action for demonstration purposes.

The demo currently contains exactly one SAP IDM form that uses Rhidmo to provide the following features:

- computing attributes using a JavaScript function on load
- validating user input using a JavaScript function on submit
- displaying validation error messages in the language of the logged in user

The recommended starting points for studying implementation details of the demo are the following designtime objects from package `de.foxysoft.rhidmo.demo.main`:

- Form **RhidmoDemoValidFromTo**
- Package script **rhidmo_onLoad_setValidFromToday**
- Package script **rhidmo_onSubmit_checkValidToNotBeforeValidFrom**

3.1 Prerequisites

You have installed and configured the Rhidmo framework as described in chapter 2.

3.2 Considerations Before Using the Demo

3.2.1 Access Control

The Rhidmo demo form ships with an access control restricting access to members of the `MX_ROLE:ADMIN` role, which exists by default in every SAP IDM installation. Users not assigned to this role will not see the demo form in their navigation menu and not be able to start it with a direct link. If your personal user doesn't have this role and it's not practical or desirable to get it, you'll not be able to use the demo as is.

3.2.2 Creation of Persistent Test Data

The most important use case of the Rhidmo demo form is demonstrating what happens when validation of user input *fails*. In this case, all data entered on the form is completely ephemeral. Everything remains in session memory and nothing will be committed to the database.

When you submit valid input though, your submit action will result in creation of a new persistent MX_PERSON entry in your database.

The rationale for providing a Create form instead of a form that modifies an existing entry is inspired by experience that creating *new* data for testing purposes is typically less problematic than modifying *existing* data. Existing data is typically there for a purpose, and we wanted to avoid making any such data inconsistent or unusable in your system.

3.2.3 Self-Deletion of Persistent Test Data

The next design aspect to carefully review is that the demo form has a result handling process which will asynchronously delete the new MX_PERSON only a couple of seconds after it has been created.

The rationale is not to pollute your database with long-lived test data and avoid any need for manual cleanup.

3.2.4 Potential Issues

While the Rhidmo demo has been designed carefully, there might be scenarios where it is not desirable or even acceptable to execute it. Potential issues to consider include:

- Custom *entry event handlers* which automatically execute whenever new MX_PERSON entries are created; these event handlers will likely fail for the entry created by the Rhidmo demo form because it may already have been deleted when the custom event handler executes. This may generate errors in your job log.
- Custom *attribute event handlers* which automatically execute whenever a new value for MX_VALIDFROM and/or MX_VALIDTO are added to the database; such handlers may fail for the same reason.
- *Organizational policy* prohibiting the deletion of SAP IDM entries, specifically of type MX_PERSON, for compliance or security reasons. As this demo creates and deletes *test data* only, this should not be an issue, but make sure to check in advance.

If you suspect that any of these potential issues could hurt in your environment, we recommend not to use the demo at all or at least not as-is. If you're a developer, you'll probably find it easy to modify the demo source code to make it more suitable for your environment, e.g. by changing the form's access control, replacing the entry type of the form with something else and/or by disabling its result handling process.

3.3 Deploy the Demo Packages

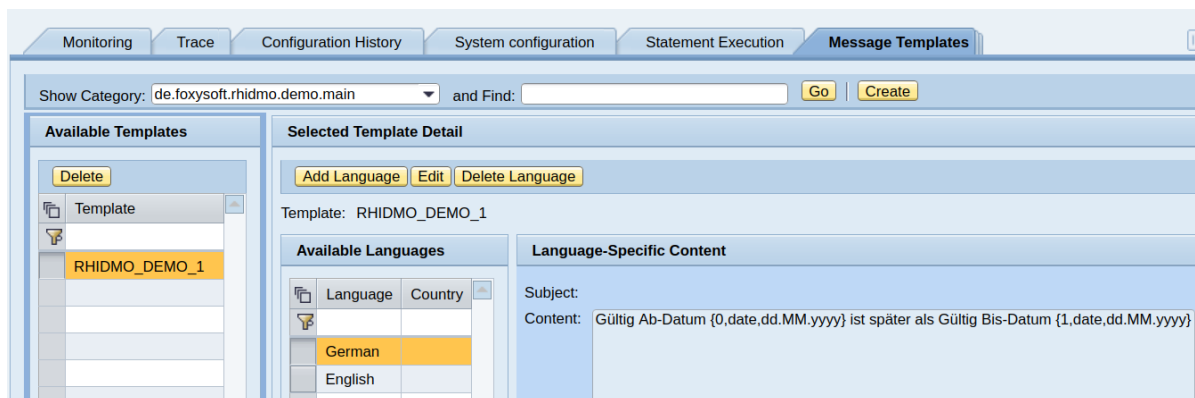
The Rhidmo demo consists of two SAP IDM packages which are located under the examples folder of the Rhidmo ZIP distribution. Unpack the ZIP distribution and use SAP IDM Developer Studio to import these package files (file extension: .idmpck) into the main identity store of your system, i.e. the identity store which the SAP IDM Web Dynpro UI operates on.

Make sure you import both demo packages into your system. The exact order of import doesn't matter.

- de.foxysoft.rhidmo.demo.util
- de.foxysoft.rhidmo.demo.main

After importing the main package, open the SAP IDM administration UI at URL path /idm/admin. On tab “Message Templates”, verify that the import has automatically created a new message template category called de.foxysoft.rhidmo.demo.main (like the package).

This message category contains language-specific versions of UI error messages used in the Rhidmo demo. Rhidmo ships with German and English language versions of these texts. Using the SAP IDM administration UI, you could add more languages as required.



3.4 Verify Your User's UI Language

All texts of the Rhidmo demo have been translated to German and English, so you may want to make sure that your UI language is set to one of those two in order to get the best demo experience. When logged in with a user whose UI language has been set to any other than those two, the UI and all message texts will use the English version as a fallback.

To check what your current UI language is, log in to SAP NetWeaver's local user administration at URL path /useradmin. Verify or change your user's UI language as illustrated below:

Welcome admin

Identity Management Import Configuration Consistency Check

Search

Search Criteria: User All Data Sources admin Go Advanced Search

Create User Copy to New User Delete Unlock Lock Generate New Password Export

Principal Type	Sta...	Logon ID	Name	Depart
		admin	admin,	

Details of User admin

Modify Save Cancel

General Information Account Information Contact Information Additional Information

Logon ID: admin

☒ Define Initial Password
☐ Generate Password
☐ Disable Password

Define Password:

Confirm Password:

Last Name: * admin
 First Name:
 E-Mail Address:
 Form of Address:
 Language: English
 Activate Accessibility Feature: ☐

Select Photo: Durchsuch
 Upload Remove

3.5 Use the Demo Form

To launch the demo form, open a browser at URL path /idm of your SAP NetWeaver host and log in with a user who is member of the MX_ROLE:ADMIN role.

Select the “Manage” tab. Keep the “Show” drop-down at its default selection “Person” and use the “Create...” button to see a list of available forms that can create new MX_PERSON entries.

A new forms folder “Rhido® Demo Forms” will be available. Expand that and launch the form “Rhido® Demo: Valid From/To with Computed Values, Validation and I18N” by using the “Choose Task” button.

The form has two mandatory attributes open for edit:

- **Valid From:** this attribute has been set to today's date automatically by package script `rhismo_onLoad_setValidFromToday`
- **Valid To:** this attribute is initially empty and must be filled in by you

You can now proceed to fill in a value for Valid To and optionally also change the computed default value of Valid From.

Since the use case is demonstrating validation *failure*, you'll want to enter a Valid From date that is *before* the Valid From date in order to see a corresponding validation error message in the UI. Validation passing is not particularly interesting, from a UI perspective.

When pressing Submit, package script `rhismo_onSubmit_checkValidToNotBeforeValidFrom` will -as its name suggests- verify that the Valid To date is not before the Valid From date. If this validation fails, the script will generate an error message that will be displayed in the UI and whose exact text depends on the UI language of your user (see chapter 3.4).

Rhidmo® Demo: Valid From/To with Computed Values, Validation and I18N

 Valid from date 6/4/2025 is after valid to date 6/3/2025

Valid From: * 
Valid To: * 

IN WHICH CASES WILL VALIDATION FAIL?
Enter a Valid To date that is before the Valid From date, then use "Submit"

If you switch your UI language as illustrated in chapter 3.4, make sure to log out and back in again to actually see the change. All your existing sessions will continue to use the previous UI language.

Rhidmo® Demo: Gültig Ab/Bis mit Berechnung, Validierung und I18N

 Gültig Ab-Datum 04.06.2025 ist später als Gültig Bis-Datum 03.06.2025

Gültig ab: * 
Gültig bis: * 

WANN KOMMT ES ZU VALIDIERUNGSFEHLERN?
Geben Sie ein Gültig Bis-Datum ein, das vor dem Gültig Ab-Datum liegt, dann drücken Sie "Senden"

For completeness sake, here's an illustration of what happens if validation succeeds: a success message will be displayed and your data be committed to the database.

Rhidmo® Demo: Valid From/To with Computed Values, Validation and I18N

 Request submitted

Valid From:
Valid To:

IN WHICH CASES WILL VALIDATION FAIL?
Enter a Valid To date that is before the Valid From date, then use "Submit"

3.6 Uninstalling the Demo

To uninstall the Rhidmo demo, use the job Uninstall Demo Message Templates located in package de.foxysoft.rhidmo.demo.main as a first step. You can either run this job from SAP IDM Developer Studio, or use the administration UI at /idm/admin > System configuration > Packages > de.foxysoft.rhidmo.demo.main.

On the “Jobs” tab, the job can be executed directly using the “Run Now” button.

The screenshot displays the SAP IDM Developer Studio administration interface. The top navigation bar includes tabs for Monitoring, Trace, Configuration History, System configuration (selected), Statement Execution, and Message Templates. On the left, a sidebar shows 'Repositories' and 'Packages'. The main content area is titled 'de.foxysoft.rhidmo.demo.main' and contains a table with the following data:

Qualified Name	Identity Store	Version
de.foxysoft.rhidmo.demo.main	FOXYSOFT_DEV	1.0

Below the table, the 'Details of Package "de.foxysoft.rhidmo.demo.main":' section is visible, with the 'Jobs' tab selected. This section shows the configuration for the 'Uninstall Demo Message Templates' job. The 'Configure and Execute Job "Uninstall Demo Message Templates"' panel includes buttons for 'Run Now', 'Stop', 'Enforce Restart', 'Save', and 'Refresh'. The job is 'Enabled' and has a 'Schedule Rule' of 'On demand'. The 'Overall Status' is 'Idle', with 'Job State: Idle', 'Dispatcher Assigned: [checked]', and 'Next Scheduled Run: Not scheduled'. The 'Display Job Runs' section shows a table with the following data:

State	Operations	Errors	Warnings	Start Time
[Green icon]	1	0	0	2025-06-05 11:49:46

After job execution, first perform a browser refresh to reload any and all content displayed in the admin UI, then switch to tab “Message Templates” to verify that message template category de.foxysoft.rhidmo.demo.main including all of its templates have been removed.

As a final step, use SAP IDM Developer Studio to delete the demo packages:

- de.foxysoft.rhidmo.demo.util
- de.foxysoft.rhidmo.demo.main

4 Developing Forms Using Rhidmo

Each form that needs to use Rhidmo must be configured. Please enter the following values in the form properties:

The screenshot shows the 'Rhidmo8TestForm' configuration window with the 'General' tab selected. The 'Enabled' checkbox is checked. The 'Form ID/Name' field contains '186/Rhidmo8TestForm'. The 'Form Type' dropdown is set to 'Regular'. The 'Repository' dropdown is set to '-- None --'. The 'Use Inactive Entries' and 'Use Context Variables' checkboxes are unchecked. The 'Extension Framework' dropdown is empty. The 'Extension Class' field contains 'de.foxysoft.rhidmo.TaskProcessingDynamic'. The 'onLoad' and 'onSubmit' checkboxes are checked. The 'Custom Parameters' table has two rows: 'ONLOAD' with type 'String' and value 'rhidmo8_onload', and 'ONSUBMIT' with type 'String' and value 'rhidmo8_onsubmit'. There are 'Add', 'Modify...', and 'Remove' buttons on the right.

Property	Type	Value
ONLOAD	String	rhidmo8_onload
ONSUBMIT	String	rhidmo8_onsubmit

Use the extension class “de.foxysoft.rhidmo.TaskProcessingDynamic”. The “ONLOAD” and “ONSUBMIT” parameters denote the JavaScript functions to be called during the corresponding events. Starting with Rhidmo release 1.1.0, you can use scripts from the same package as the form and/or scripts from any other package.

For scripts from the same package as the form, use the following **simple syntax** for the value of properties:

`<script_name>`

For scripts from other packages, use the following **extended syntax**:

`<package_name>/<script_name>`

If your ONLOAD/ONSUBMIT script depends on other scripts, you need to specify these dependencies using properties REQ1 to REQn, where 1..n is an interval of contiguous integers which must start at 1.

For instance, if your ONLOAD script which has 3 dependencies, you need to specify:

REQ1=<first dependency>

REQ2=<second dependency>

REQ3=<third dependency>

The following screenshot illustrates this. The main script ONLOAD has three dependencies. Two of these, REQ1 and REQ2, use the simple syntax (script name only). The third dependency, REQ3, uses the extended syntax to refer to a script from package de.foxysoft.bobj.

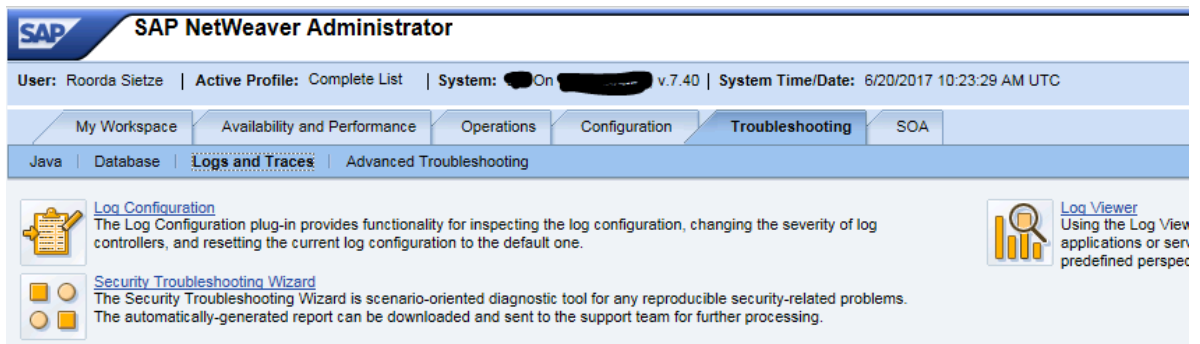
The screenshot shows a configuration window for an extension framework. At the top, there is a label 'Extension Framework' followed by a text input field. Below this, the 'Extension Class' is set to 'de.foxysoft.rhidmo.TaskProcessingDynamic'. To the right of the extension class, there are two checkboxes: 'onLoad' (checked) and 'onSubmit' (unchecked). Under the 'Custom Parameters:' label, there is a table with three columns: 'Property', 'Type', and 'Value'. The table contains four rows of data. To the right of the table, there are three buttons: 'Add', 'Modify', and 'Remove'.

Property	Type	Value
ONLOAD	String	rhidmo8_onload
REQ1	String	fx_trace
REQ2	String	fx_getConstant
REQ3	String	de.foxysoft.bobj/fx_JavaUtils

5 Rhidmo Troubleshooting

Please check the JAVA AS log in case something goes wrong. If necessary, make sure that the log configuration shows the right amount of messages:

The log viewer can be configured and accessed using the “/nwa” application (go to “Troubleshooting” → “Logs and Traces”):



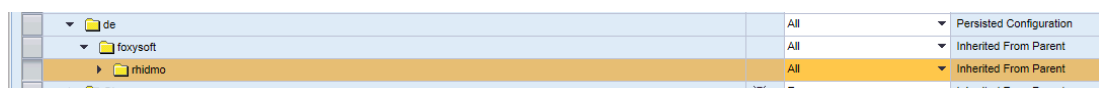
5.1 Configuration

By clicking on the link “Log Configuration” you can configure the amount of logging that Rhidmo generates:

1. Select “Tracing Locations”:



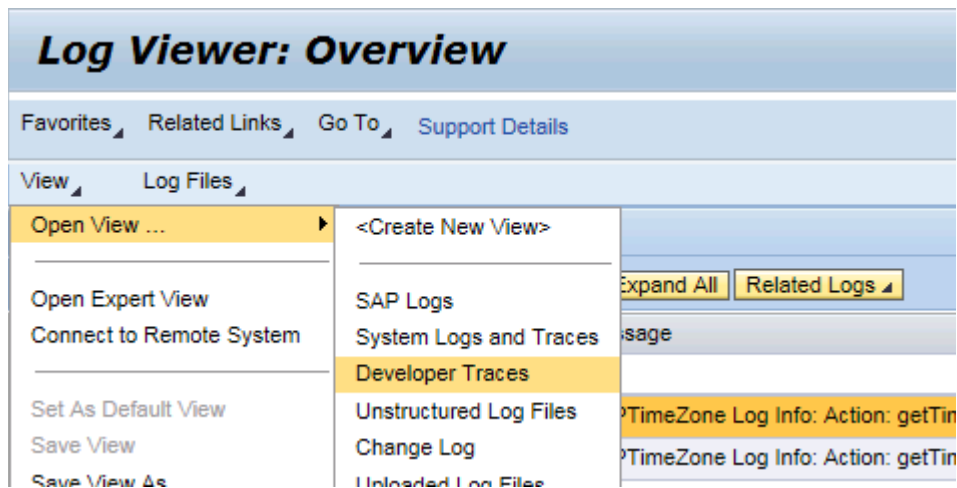
2. Then select “de.foxysoft.rhidmo” in the subtree below:



3. Select the log level you want from the drop-down box to the right and click on the “Copy to Subtree” button to copy this setting to all locations below.

5.2 Log Viewer

Click on the “Log Viewer” link to view the logs (Select the View “Developer Traces” to examine the Rhidmo logs):



Example log:

Developer Traces					
Show Advanced Filter [Icons] Expand All Related Logs					
Detail	Severity	Date	Time	Message	Location
		▼ yyyy-mm-dd			
+	debug	2017-06-20	13:37:08.275	service: Returning [com.sap.idm.extension.apl.idmValue@4637afb]	de.foxysoft.rhidmo.TaskProcessingStatic
+	warning	2017-06-20	13:37:08.256	Attribute [MX_PHONE_PRIMARY] Value []	de.foxysoft.rhidmo.Application
+	warning	2017-06-20	13:37:08.256	Attribute [DISPLAYNAME] Value [Test SBR]	de.foxysoft.rhidmo.Application
+	warning	2017-06-20	13:37:08.256	Attribute [MSKEYVALUE] Value [TESTSBR]	de.foxysoft.rhidmo.Application
+	warning	2017-06-20	13:37:08.256	OnLoad called. Subject MSKEY [23] object MSKEY [25]	de.foxysoft.rhidmo.Application
+	debug	2017-06-20	13:37:08.256	service: f = [org.mozilla.javascript.gen.rhidmo8_onload_23@362f786d]	de.foxysoft.rhidmo.TaskProcessingStatic

6 FAQ

6.1 “Invalid Key size”

The following two messages appear in the log file and (en-) or decryption is not possible:

	 error	2018-03-18	10:23:41:130	java.security.InvalidKeyException: Illegal key size at javax.crypto.Cipher.a(DashoA13*...) at javax.crypto.Cipher.a(DashoA13*...) at javax.crypto.Cipher.a(DashoA13*...) at javax.crypto.Cipher.a(DashoA13*...)
	 debug	2018-03-18	10:23:41:130	determineSecretKey: Keysize = 32 Maximum 16

The second message tells us that maximum supported key size is 16 bytes (or 128 bits). However, the string is encrypted with a key having 32 bytes (or 256 bits).

This means that the SAP JAVA AS is running with a Java version with limited cryptography capabilities. See SAP Note 1240081 for further details.